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Generation and Validation of Synthetic
Smart Home Sensor Data
for Detecting Behavioural Drift in
Early-Stage Alzheimer's Care

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A report submitted in partial fulfilment of the requirements of Aston
University for the degree of Master of Science in *Artificial Intelligence with
Business Strategy*

September 12, 2025

Declaration

I, Sharanya Panicker, of the College of Engineering and Physical Sciences, Aston University, confirm that this dissertation is my own work. All figures, tables, equations, code snippets, artworks, and illustrations included in this report are original and have not been taken from any other person's work, except where the works of others have been explicitly acknowledged, quoted, and referenced. I understand that failure to do so will be considered a case of plagiarism. Plagiarism is a form of academic misconduct and will be penalised accordingly.

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Sharanya Panicker

September 12, 2025

Abstract

Monitoring the daily lives of older adults with dementia without intruding on their privacy is a critical area of research, particularly for smart-home technology designed to detect unusual behaviour. However, we've hit a roadblock: it's challenging to obtain high-quality datasets that demonstrate how a person's behaviour gradually changes over time. Gathering this kind of information in the real world is an ethically complex process, often resulting in incomplete data. To solve this problem, this dissertation was conducted in three phases. First, synthetic datasets were created using the SHARON, configured to mimic Activities of Daily Living (ADLs) and then added irregularities and staged behavioural changes that are typical of dementia. Then, the realism of these datasets is evaluated by comparing them against the CASAS benchmark, using entropy and Jaccard similarity. This provided a way to examine both the overall behavioural diversity and how closely the generated data matched the real-world benchmark. Finally, trained four anomaly-detection models—Isolation Forest, One-Class SVM, an Autoencoder, and an LSTM sequence predictor on the initial, pre-drift data. Their performance was tested after training them on different data durations: 30, 90, and 120 days. It was concluded that SHARON-generated data is quite similar to real data. The entropy analysis revealed more well-crafted synthetic sequences compared to the CASAS benchmark, while the Jaccard similarity analysis indicated a significant overlap and few differences in the activities. Out of all the models tested, the deep learning approaches, especially the Autoencoder, performed the best overall (achieving an F1 score up to 0.95). In contrast, the more traditional models tended to have higher precision but were less effective at adapting to behavioural drift. It was also found that increasing the training duration improved the models' recall and stability, though the gains became less significant after 90 days. This study provides a repeatable process for creating and validating synthetic smart-home data that includes behavioural drift. It also highlights the limitations of current simulators and provides suggestions for creating more realistic datasets that can better support dementia care applications.

Keywords: behavioural drift; synthetic smart-home data; dementia; anomaly detection; unsupervised learning.

Acknowledgements

I am deeply grateful to my supervisor, Dr. Martin Rudorfer, for his expert guidance and invaluable feedback throughout this project. His insightful comments were instrumental in shaping the direction of my research. I would also like to thank my peers for their support and my family for their unwavering encouragement during this demanding period. Finally, I extend my thanks to the developers of the SHARON and CASAS projects for making their publicly available resources a cornerstone of this study. All remaining errors are, of course, my own.

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List of Abbreviations

ADLs	Activities of Daily Living
AUC	Area Under the Curve
CASAS	Center for Advanced Studies in Adaptive Systems (real-world smart-home dataset)
IF	Isolation Forest
LSTM	Long Short-Term Memory (neural network architecture)
MSE	Mean Squared Error
OCSVM	One-Class Support Vector Machine
PR-AUC	Precision–Recall Area Under the Curve
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SHARON	Simulator of Human Activities, ROutines and Needs

Chapter 1

Introduction

1.1 Context and Motivation

The world's population is ageing rapidly, with those aged 65 and above projected to rise from 10% in 2022 to 16% by 2050 (Wang et al., 2023). In Europe, this cohort is expected to exceed one-third of the population by mid-century (Deep et al., 2020), while the WHO similarly anticipates a doubling of the global share of people aged 60 and over within the same period (Cejudo et al., 2025). Ageing is closely associated with chronic illness, cognitive decline, and memory impairment, all of which account for a substantial proportion of healthcare expenditure in high-income countries (Deep et al., 2020; Cejudo et al., 2025). At the same time, shortages of long-term care workers continue to strain services (Tanaka et al., 2024). These pressures underscore the need for innovative approaches to support independent living, with smart-home monitoring systems increasingly recognised as a promising technological response.

Within this context, dementia, and particularly Alzheimer's disease, presents a major and urgent global challenge. In 2021, approximately 57 million people worldwide had dementia, with nearly 10 million new cases each year, making it currently the seventh leading cause of death and a major cause of disability and dependency globally (WHO, 2025). Alzheimer's disease is the most common form, contributing to an estimated 60–70% of dementia cases (WHO, 2025). While mainly related to ageing, dementia is not an inevitable component of ageing, and younger-onset cases (before age 65) do amount to up to 9% worldwide, with an estimated 200,000 younger-onset instances just in the United States (Alzheimer's Association, 2019). As illustrated in Figure 1.1, the scale of the condition is striking. In the United States, 5.8 million people were living with Alzheimer's dementia in 2019, of whom 81% were aged 75 or older; this number is projected to rise to 13.8 million by 2050 (Alzheimer's Association, 2019). At the global level, prevalence is expected to triple by 2050, with nearly 68% of new cases emerging in

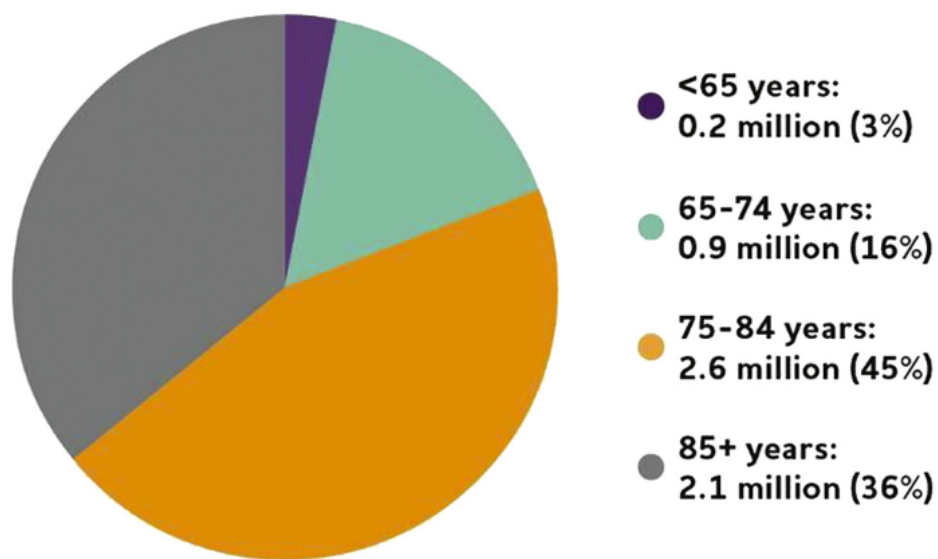


Figure 1.1: Age distribution of people with Alzheimer's dementia in the United States, 2019.

Source: Alzheimer's Association (2019, p. 330).

low- and middle-income countries, as shown in Figure 1.2 (Lentzas and Vrakas, 2020; Alzheimer's Association, 2019).

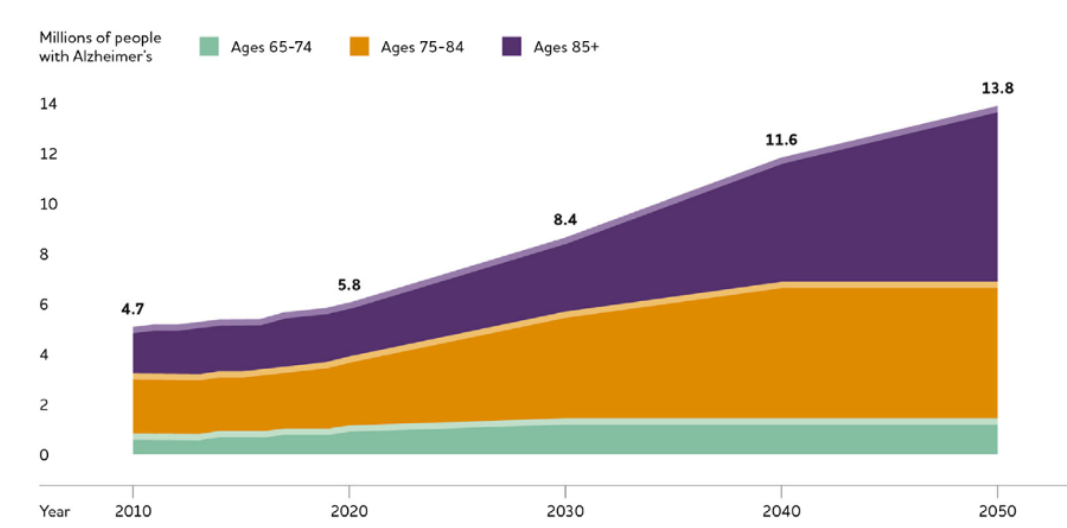


Figure 1.2: Projected number of people aged ≥ 65 years with Alzheimer's dementia in the United States, by age group, 2010–2050.

Source: Alzheimer's Association (2019, p. 335).

Dementia presents a dramatic economic challenge, costing the global economy \$1.3 trillion in 2019, with projections of a two-fold increase by 2050 (WHO, 2021). The burden is particularly stark in the U.S., where total care costs reached \$290 billion in

2019 and are projected to surpass \$1.1 trillion by 2050 (Alzheimer's Association, 2019). The financial strain on individuals is significant, with annual per-person payments for Medicare beneficiaries with dementia nearly three times higher than for those without (\$48,977 vs. \$13,976 in 2018). As shown in Figure 1.3, Medicare and Medicaid account for the majority of these payments. The challenges are not limited to economics; dementia also severely interferes with activities of daily living and places a considerable burden on informal caregivers. More than 16 million Americans provided an estimated 18.5 billion hours of unpaid care in 2018, with a significant majority of caregivers reporting high levels of emotional stress (59%) and physical strain (38%), as illustrated in Figure 2.3 (Alzheimer's Association, 2019). These multifaceted challenges are why the World Health Organisation designated dementia a global public health priority (WHO, 2019).

Payment source	Beneficiaries with Alzheimer's or other dementias	Beneficiaries without Alzheimer's or other dementias
Medicare	\$24,598	\$7561
Medicaid	8565	365
Uncompensated	381	382
Health maintenance organization	1261	1544
Private insurance	2253	1422
Other payer	937	242
Out of pocket	10,798	2336
Total*	48,977	13,976

Figure 1.3: Average annual per-person payments by payment source (Medicare Beneficiaries Aged ≥ 65 Years), with and without Alzheimer's or other dementias, 2018 (US Dollars).

Source: Alzheimer's Association (2019, p. 349).

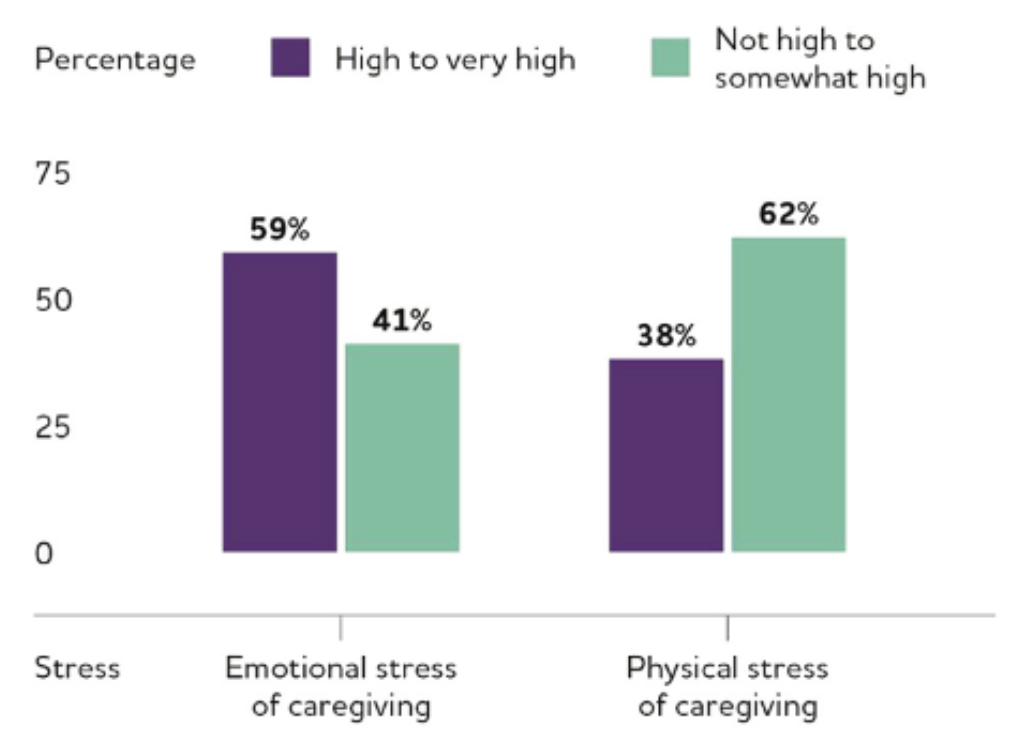


Figure 1.4: Reported emotional and physical stress levels among caregivers of people with Alzheimer's dementia, United States, 2018.

Source: Alzheimer's Association (2019, p. 344).

Rather than abrupt, isolated anomalies, dementia often manifests through behavioural drift: gradual, systematic changes to daily routines such as delayed or fragmented sleep, meal skipping, or repeated toileting. Such behavioural drifts indicate increased challenges in performing simple life tasks or even the onset of illnesses such as Alzheimer’s or dementia (Azefack et al., 2019; Belmonte-Fernández et al., 2020). Existing simulation approaches typically rely on random or ad hoc anomaly injection, which does not accurately reproduce the context-dependent, progressive nature of dementia-related changes (Ordóñez et al., 2015; Francillette et al., 2020; Tanaka et al., 2024). Identifying subtle yet progressive shifts is essential for early intervention; however, current technologies for analysis and datasets appear limited in their capacity to capture these trajectories. Collecting ground-truth data reflecting dementia-related drift is both ethically sensitive and expensive, often being incomplete, with annotations prone to recall bias or privacy constraints. Established datasets such as CASAS provide valuable baselines, but they predominantly capture routine behaviours, limiting their utility for benchmarking detectors of progressive change.

Chapter 2

Literature Review

2.1 Dementia and Its Impact

Dementia, a progressive brain disorder, represents one of the most urgent and complex challenges facing global health systems (WHO, 1994; Alzheimer's Association, 2019). While predominantly associated with ageing, it is not an inevitable outcome of growing older, and approximately 200,000 younger-onset cases are reported in the United States alone (Alzheimer's Association, 2019). Alzheimer's disease is the most prevalent form, accounting for an estimated 60–70% of dementia cases (WHO, 2025).

The scale of the condition is striking. In the United States, 5.8 million people were living with Alzheimer's dementia in 2019, of whom 81% were aged 75 or older (Alzheimer's Association, 2019). Figure 2.1 illustrates the strong age-related distribution of prevalence. This number is projected to rise to 7.1 million by 2025 and 13.8 million by 2050, largely driven by demographic ageing (Alzheimer's Association, 2019). Figure 2.2 visualises this rapid upward trajectory in U.S. prevalence. Globally, dementia prevalence is expected to triple by 2050, with nearly 68% of new cases emerging in low- and middle-income countries (Lentzas & Vrakas, 2020; Alzheimer's Association, 2019).

The economic burden is equally severe. In the U.S., total care costs reached \$290 billion in 2019 and are projected to surpass \$1.1 trillion by 2050 (Alzheimer's Association, 2019). On a per-person basis, annual healthcare and long-term care costs for people with dementia average \$48,977, nearly three times higher than for those without dementia (\$13,976). Table 2.1 summarises these per-capita costs.

Beyond financial strain, dementia profoundly disrupts daily life and places a disproportionate burden on informal caregivers. In 2018, over 16 million Americans provided 18.5 billion hours of unpaid care, with 59% reporting high emotional stress and 38% reporting physical strain (Alzheimer's Association, 2019). Figure 2.3 illustrates these caregiver stress levels.

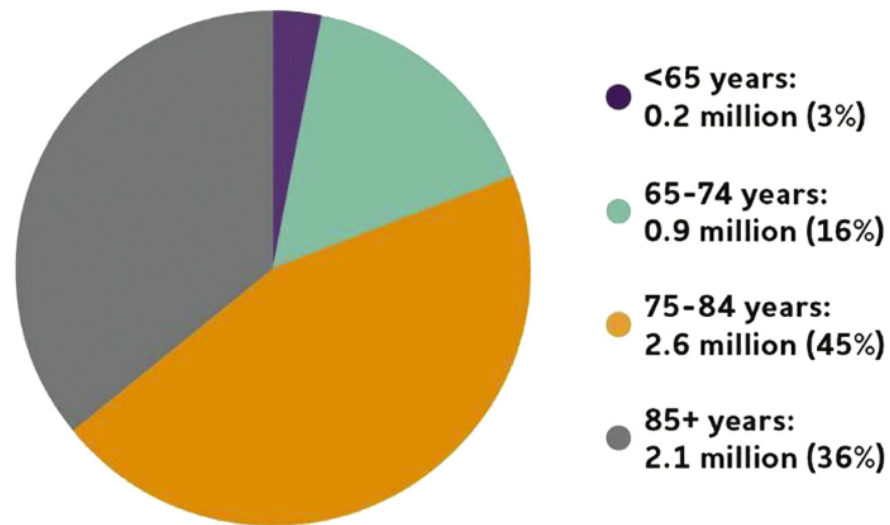


Figure 2.1: Age distribution of people with Alzheimer's dementia in the United States, 2019. *Source:* Alzheimer's Association (2019, p. 330).

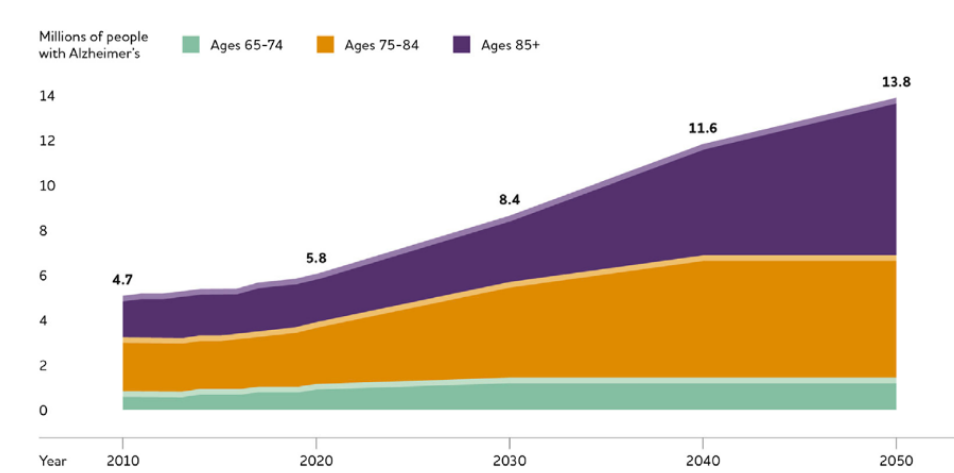


Figure 2.2: Projected number of people aged ≥ 65 years with Alzheimer's dementia in the United States, by age group, 2010–2050. *Source:* Alzheimer's Association (2019, p. 335).

Table 2.1: Average annual per-person payments by payment source (Medicare beneficiaries aged ≥ 65 years), with and without Alzheimer's or other dementias, 2018; US dollars. *Source:* Alzheimer's Association (2019, p. 349).

Payment source	With Alzheimer's or other dementias	Without Alzheimer's or other dementias
Medicare	24,598	7,561
Medicaid	8,565	365
Uncompensated	381	382
Health maintenance organization	1,261	1,544
Private insurance	2,253	1,422
Other payer	937	242
Out of pocket	10,798	2,336
Total	48,977	13,976

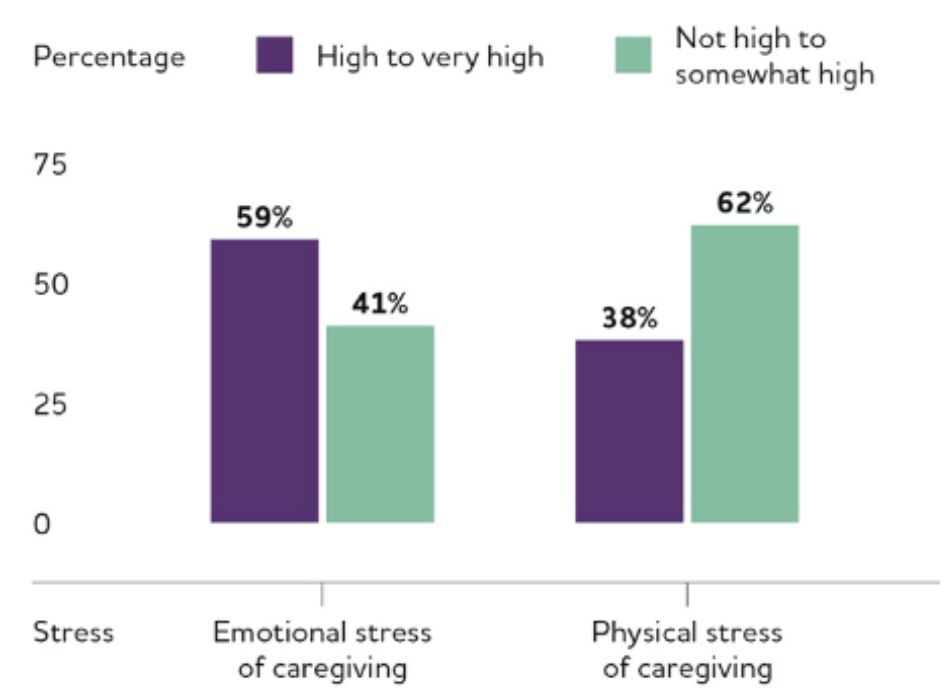


Figure 2.3: Reported emotional and physical stress levels among caregivers of people with Alzheimer's dementia, United States, 2018. *Source:* Alzheimer's Association (2019, p. 344).

Such challenges underscore why dementia has been designated a global public health priority by the World Health Organisation (WHO, 2019). Against this backdrop, there is growing recognition that traditional models of care are unsustainable. As populations age and workforce shortages intensify (Tanaka et al., 2024), technology-enabled interventions, such as unobtrusive smart-home monitoring, are increasingly seen as essential. By capturing behavioural patterns and detecting subtle anomalies, these systems offer potential to reduce reliance on costly reactive care and to support early intervention strategies.

2.2 Smart Homes as a Solution

Smart homes are increasingly promoted as unobtrusive monitoring systems to address the dual challenges of population ageing and rising care costs (Wang et al., 2023). They align with the widespread preference of older adults to age in place by supporting independence and enhancing quality of life (Jakkula & Cook, 2011; Deep et al., 2020). Compared with wearables (adherence/battery) and vision-based systems (privacy/lighting), ambient sensors provide a practical solution (Deep et al., 2020; Tanaka et al., 2024). Passive infrared detectors, door sensors, and pressure mats are inexpensive, minimally invasive, and capable of constructing detailed temporal profiles of daily routines (Wang et al., 2023; Tanaka et al., 2024). Large-scale initiatives such as CASAS demonstrate that activity levels, sleep quality, and task engagement can be tracked to reveal both short-term well-being and long-term behavioural trends (Cook et al., 2013; Cejudo et al., 2025).

2.3 Abnormal Behaviour Detection in Smart Homes

Smart home systems play a crucial role in detecting abnormal behaviours, defined as deviations from expected daily routines and often serving as early markers of cognitive impairment such as Alzheimer’s disease (Wang et al., 2023; Deep et al., 2020). These deviations are not always abrupt but can emerge gradually as subtle behavioural shifts preceding clinical diagnosis. Traditional anomaly categories, such as duration-based, order-based, or point anomalies, offer applicable classification schemes but are limited in flexibility. Dementia-related drift rarely manifests as discrete categories; instead, it unfolds as overlapping and evolving patterns. For example, disrupted sleep may simultaneously involve temporal anomalies (late onset) and order anomalies (nighttime awakenings), illustrating the insufficiency of rigid classifications for progressive neurodegenerative conditions.

Evidence increasingly confirms that dementia-related anomalies evolve in both frequency and severity, making them valuable early indicators of decline (Wang et al., 2023;

Tanaka et al., 2024). Longitudinal studies reveal measurable activity changes even in early stages, such as reduced moderate-intensity movement, diminished morning activity complexity, and lower overall peaks (Varma & Watts, 2017). Such findings suggest that anomalies should be conceptualised not as isolated outliers but as trajectories of gradual behavioural change. This reconceptualisation challenges anomaly detection systems to move beyond binary classification and incorporate both progressive drift and sudden hazardous events.

To address this complexity, researchers employ a diverse range of methods. Probabilistic approaches such as cumulative distribution functions (CDFs) delineate behavioural boundaries (Wang et al., 2023). Conventional classifiers, including Support Vector Machines (SVMs), perform reasonably well in binary contexts but depend on handcrafted features and lack adaptability to evolving drift. Deep learning techniques, particularly hybrid CNN–LSTM architectures, show promise for modelling temporal dependencies (Wang et al., 2023), yet high data requirements constrain their effectiveness, which is rarely met in dementia-focused research (Tanaka et al., 2024). This highlights a persistent trade-off: interpretable but limited classical models versus data-hungry yet powerful deep models.

Anomaly detection must also address acute risks such as falls, missed meals, or nocturnal wandering. While some studies broaden coverage by analysing dozens of Activities of Daily Living (ADLs) to improve interpretability (Cejudo et al., 2025), many frameworks remain generic to “elderly care” rather than tailored to dementia-specific trajectories. This creates a misalignment: whereas clinical guidelines from the WHO (2019,2025) and NHS (2024) emphasise structured routines and carer support, technical models often fail to capture dementia-specific anomalies such as repetitive toileting or disoriented wandering.

In summary, despite advances in anomaly detection, current approaches remain limited. Taxonomies oversimplify behavioural drift, classical models lack adaptability, and deep models face data scarcity. More critically, few systems explicitly account for the progressive and overlapping nature of dementia-related anomalies. These shortcomings underscore the value of synthetic data generation, which provides a means to model progressive decline and test detection methods in a controlled yet clinically relevant context.

2.4 The Challenge of Data Scarcity and the Rise of Synthetic Data

Yet even as detection methods advance, their evaluation remains hampered by a persistent bottleneck: the scarcity of suitable datasets. Widely used corpora such as CASAS (Cook et al., 2013), Kyoto, and MavHome provide valuable sensor traces of daily living in controlled environments but rarely capture dementia-related anomalies. Behaviours such as nocturnal wandering, repeated toileting, or progressive meal-skipping are infrequent and ethically difficult to document, resulting in datasets dominated by routine activities (Tanaka et al., 2024; Wang et al., 2023). As a result, models trained on these resources are often ill-equipped to detect the very behaviours most critical for early intervention.

The collection of real-world anomaly data also poses significant ethical and logistical challenges. Continuous monitoring of vulnerable populations raises privacy concerns and risks of intrusiveness for both patients and caregivers (WHO, 2019; NHS, 2024). Policy frameworks emphasise autonomy and dignity while supporting assistive technologies, yet balancing these priorities in practice is difficult. Moreover, dementia-related anomalies are unpredictable, making long-term, large-scale data collection costly and complex. Even where data exist, annotation remains problematic: validated ground truth is rare, and retrospective caregiver reports are subject to recall bias (Deep et al., 2020). Thus, a paradox emerges: robust detection systems require rich labelled datasets, but the groups most in need are the hardest to study.

In response, researchers increasingly turn to synthetic data as a complement to empirical observation (Tanaka et al., 2024; Veronese et al., 2015). Simulation offers distinct advantages: it enables large-scale data generation under controlled conditions (Tanaka et al., 2024), systematic modelling of anomaly types (Arifoglu & Bouchachia, 2019), and avoidance of privacy risks (Deep et al., 2020; Xu et al., 2025). Crucially, it allows safe experimentation with rare or hazardous events, such as falls or prolonged wandering, that cannot ethically be staged. Beyond isolated events, simulators can model progressive behavioural drift, such as deteriorating sleep patterns, reflecting the longitudinal dynamics of dementia (Tanaka et al., 2024).

Several frameworks support these aims. SHARON generates daily living routines using parameterised activities and environmental constraints (Veronese et al., 2015) but has been criticised for emphasising routine patterns over progressive or interdependent anomalies (Tanaka et al., 2024). More recent systems, including OpenSHS, offer configurable anomaly injection, while others integrate latent variables to represent cognitive decline, better reflecting Alzheimer’s trajectories (Tanaka et al., 2024). Yet many still rely on simplistic strategies, such as randomly inserting sensor events or fabricating abnormal situations,

which lack ecological validity. In reality, dementia-related disruptions accumulate and interact, producing overlapping anomalies that current simulators often fail to replicate.

Against this backdrop, this dissertation contributes by employing SHARON to generate synthetic smart-home data and validating its realism using entropy and Jaccard similarity. This dual focus addresses both data scarcity and ecological validity, providing a systematic basis for evaluating anomaly detection models in dementia care.

2.5 Simulation Frameworks and Their Capabilities

Multiple simulation frameworks have been developed to generate synthetic smart-home datasets, each varying in realism and flexibility. The SHARON simulator reproduces daily routines through parameterised activities, needs, and environmental constraints, producing long sequences of sensor activations suitable for anomaly detection research (Veronese et al., 2015). Its key strengths are efficiency, stability, and routine customisation. However, its focus on replicating typical activity patterns limits its capacity to model the progressive and interdependent anomalies associated with dementia (Tanaka et al., 2024).

In contrast, OpenSHS has been applied in the creation of the SIMADL dataset, where researchers simulated anomalies such as leaving appliances or doors open for extended periods (Alshammari et al., as cited in Tanaka et al., 2024). While useful for anomaly injection, these implementations often relied on manual trace modifications, and detailed accounts of the framework’s architectural flexibility remain limited, restricting evaluation of its broader applicability. More recently, Tanaka et al. (2024) proposed a simulator explicitly designed to capture dementia-related decline. By incorporating latent variables such as cognitive function scores, it dynamically adjusts the frequency and severity of behaviours like wandering, sleep disruption, and forgetfulness. This approach better reflects the longitudinal and compounding character of dementia-related anomalies.

Collectively, these frameworks illustrate a shift from general-purpose activity simulation and ad hoc anomaly injection toward specialised tools capable of modelling progressive behavioural drift. Nevertheless, SHARON remains an appropriate foundation for this dissertation. Its maturity, open-source accessibility, and stability provide a practical platform for extending its routines to incorporate dementia-specific drift and anomalies.

2.6 Validating Synthetic Smart-Home Data: Entropy and Jaccard

While frameworks such as SHARON, OpenSHS, and more recent anomaly-focused simulators have advanced synthetic data generation, their value depends on how closely simulated traces mirror real behavioural and sensor patterns. The literature consistently warns that synthetic data can diverge from lived routines, making validation a critical step (Wang et al., 2023; Cejudo et al., 2025; Tanaka et al., 2024). To address this, recent studies have introduced similarity-based evaluations, emphasising the need for transparent and quantitative metrics to establish dataset credibility before use in anomaly detection research.

This dissertation adopts two complementary validation measures: entropy and Jaccard similarity. Entropy has been widely used in smart-home research to quantify variability and predictability in activity or sensor-event distributions (Lentzas & Vrakas, 2020; Wang et al., 2023). Higher entropy values indicate more diverse and complex routines, whereas lower values reflect repetitive behaviour. By applying entropy to both per-sensor firing distributions and activity–time allocations, synthetic datasets can be benchmarked against real-world baselines such as CASAS, offering insight into whether they capture realistic behavioural richness (Cook et al., 2013). Jaccard similarity, by contrast, provides a straightforward way of assessing the overlap between sets of activities or active sensors across datasets. It measures the degree to which synthetic repertoires align with real-world ones, thereby addressing not only behavioural complexity but also the breadth of daily activities represented (Jaccard, 1901; Verma & Aggarwal, 2020).

Taken together, these measures provide complementary perspectives: entropy captures internal behavioural diversity, while Jaccard evaluates external comparability. Combined with sequence-level checks such as Levenshtein distance used in recent simulators (Tanaka et al., 2024), they form a concise yet rigorous validation framework. In this dissertation, their application ensures that SHARON-generated data demonstrates sufficient realism to support the reliable evaluation of anomaly detection models.

2.7 Anomaly Detection Models in Smart Homes

Validating synthetic datasets is only meaningful if they can be applied to the training and benchmarking of anomaly detection models. Within the smart-home literature, two dominant strands emerge: classical machine learning approaches and deep learning methods. Classical approaches such as Isolation Forests (Liu et al., 2008) and One-Class SVMs (commonly attributed to Schölkopf et al., 2001, cited in Jakkula & Cook, 2011) are

widely adopted in unsupervised contexts where labelled anomalies are scarce. Isolation Forests partition the feature space to isolate rare observations, whereas One-Class SVMs define decision boundaries around “normal” behaviour. While effective on structured features derived from sensor logs, these methods struggle to capture the sequential and contextual dependencies that characterise daily routines (Deep et al., 2020; Wang et al., 2023).

By contrast, deep learning methods such as Autoencoders and Long Short-Term Memory (LSTM) networks are designed to model temporal dynamics. Autoencoders detect anomalies through elevated reconstruction errors, while LSTMs capture long-range dependencies in activity sequences (Malhotra et al., 2015; Tanaka et al., 2024). These approaches are well suited to progressive or context-dependent anomalies, such as repeated toileting or disrupted sleep, but demand large and diverse datasets to generalise effectively.

Overall, the reviewed studies highlight a fundamental trade-off: while classical models offer efficiency in low-data contexts, they struggle to capture temporal dependencies, whereas deep learning methods provide greater expressive power but demand substantially larger datasets. The persistent scarcity of annotated real-world datasets constrains both approaches, reinforcing the value of simulation frameworks that can generate realistic synthetic data. For this dissertation, such frameworks provide the foundation to systematically evaluate anomaly detection across both classical and deep learning paradigms.

2.8 Research Gaps and Rationale

While the literature demonstrates notable progress in both modelling techniques and simulation frameworks, several critical challenges persist. Traditional anomaly categories, such as duration-based, order-based, or point anomalies, offer applicable classification schemes but are often limited in their flexibility to capture the progressive and overlapping nature of dementia-related behavioural drift. This is a complex problem, as disrupted sleep, for example, may simultaneously involve temporal anomalies (late onset) and order anomalies (nighttime awakenings), illustrating the insufficiency of rigid classifications for progressive neurodegenerative conditions.

Recent advancements have begun to address this complexity. For instance, Tanaka et al. (2024) explicitly highlight the insufficiency of previous simulators that “do not consider the facts that 1) there exists a correlation between anomalies and 2) anomalies increase and/or worse over time”. Their proposed simulator specifically aims to overcome these limitations by modeling six typical anomalies “governed in common by a single latent variable associated to the degree of dementia” (Tanaka et al., 2024). This approach

dynamically adjusts the frequency and severity of behaviours like wandering, sleep disruption, and forgetfulness, reflecting the longitudinal and compounding character of dementia-related anomalies (Tanaka et al., 2024). Similarly, Belmonte-Fernández et al. (2020) introduced a novel method for detecting "linear drift in ADL modelled as circular normal distributions," specifically identifying its occurrence in "cognitive decline, Alzheimer's disease or dementia". Their work draws on statistical process control techniques to detect and estimate the change point in time when such a linear drift starts (Belmonte-Fernández et al., 2020).

Despite these important advancements, comprehensive, publicly available simulation frameworks that fully capture the progressive, interdependent, and multi-faceted nature of dementia-related behavioural drift, and are also suitable for robust comparative benchmarking, remain limited. For example, while Tanaka et al.'s (2024) simulator is highly relevant, it remains a prototype not yet publicly released for broader research use. Other existing simulation tools, such as SHARON, though mature and reproducible, often emphasise routine patterns over progressive or interdependent anomalies (Tanaka et al., 2024) and may not generate transitional movement between rooms, which can contribute to lower entropy values compared to real-world datasets like CASAS. Furthermore, challenges in modelling complex interactions, such as shared hunger needs affecting multiple eating activities, can limit the realism of simulated drift. This persistent gap in ecologically valid datasets, particularly those with systematic, reproducible, and configurable behavioural drift, hinders the development and rigorous evaluation of anomaly detection systems for dementia care.

In response, this dissertation makes three contributions. First, it generates synthetic smart-home sensor data using SHARON, extended to reflect dementia-related anomalies. Second, it systematically validates the realism of this data against a real-world benchmark dataset using entropy and Jaccard similarity. Third, it applies a range of anomaly detection models to evaluate performance under controlled conditions. Collectively, these steps aim to advance the methodological foundations for simulation in dementia-focused anomaly detection research and provide a clearer pathway toward clinical translation in dementia care.

Chapter 3

Methods

3.1 Research Design

This dissertation adopts a three-stage experimental design comprising: (i) the generation of synthetic smart-home datasets, (ii) validation of their realism against a real-world benchmark, and (iii) the application of anomaly detection models for performance evaluation. This staged structure directly addresses the research questions concerning (a) the extent to which synthetic data can replicate the statistical and behavioural properties of real datasets, (b) the usefulness of such data for training and benchmarking anomaly detection models, and (c) the impact of progressive behavioural drift on model robustness. The rationale for this design stems from the limitations of existing datasets. Public datasets, such as CASAS, provide valuable baselines but rarely capture dementia-related anomalies, are logistically challenging to extend, and raise ethical concerns regarding the long-term monitoring of vulnerable groups. As the literature emphasises, anomaly detection models are highly sensitive to the quality and variety of training data (Wang et al., 2023; Tanaka et al., 2024). Without datasets that adequately represent both normal routines and progressive dementia-related drift, models risk being either overfitted to routine activity or unable to recognise context-specific anomalies.

To address this, the research combines simulation and comparative validation. Synthetic data will be generated using the SHARON simulator (Veronese et al., 2015), chosen for its stability, reproducibility, and ability to parametrise daily living activities under controlled conditions. Realism will be benchmarked against the CASAS dataset (Cook et al., 2013) using two complementary statistical measures: entropy, capturing behavioural diversity, and Jaccard similarity, assessing overlap between activity groups. This ensures the synthetic data are not abstract artefacts but empirically grounded. The final stage evaluates anomaly detection models spanning both classical machine learning and deep learning paradigms. Isolation Forests, One-Class SVMs, Autoencoders, and LSTM

networks are selected to represent the trade-off between interpretability and expressiveness. Their performance will be assessed across synthetic datasets with and without behavioural drift, providing controlled insights into model adaptability, scalability, and robustness.

The design emphasises rigour, reproducibility, and relevance. Rigour is achieved through systematic validation and benchmarking; reproducibility is supported by open-source tools (SHARON), public datasets (CASAS), and reproducible code developed in Python and Java; and relevance is ensured by aligning simulated anomalies with dementia-specific behaviours identified in clinical literature. The project will also produce practical artefacts, including annotated datasets, simulation configurations, and source code, which serve both as evidence of practical realisation and as resources for future research. Collectively, this design provides a coherent pathway from data generation to model evaluation, directly addressing the gaps identified in the literature and advancing the methodological foundations for dementia-focused anomaly detection research.

3.2 Synthetic Data Generation

The core dataset for this study was generated using the SHARON simulator (Veronese et al., 2015), a Java-based framework designed to reproduce daily living activities through parameterised routines. SHARON was selected because it provides an open-source and reproducible platform, with a clear architecture for modelling activity–need interactions and temporal dependencies. Unlike more prototype-based simulators that focus narrowly on anomaly injection, SHARON enables the generation of long, continuous sequences of ambient-sensor activations under controlled conditions, making it a suitable foundation for adaptation to dementia-related contexts. Alternative frameworks, such as OpenSHS or Tanaka et al.’s anomaly-focused simulator, were considered but found less practical: OpenSHS offers limited documentation and primarily manual anomaly injection (Tanaka et al., 2024), while Tanaka’s simulator remains a prototype not publicly released. SHARON therefore provides the most appropriate foundation for extending simulation to dementia-related contexts.

Daily routines will be designed to reflect the Activities of Daily Living (ADLs) typical of older adults living independently. These include essential activities such as sleeping, eating (breakfast, lunch, dinner), toileting, showering, and social or leisure activities (e.g., watching television). SHARON encodes these routines through parameter sets: needs (e.g., hunger, tiredness), growth rates (γ_n), activity weights ($w_{n,a}$), effects ($\zeta(a)$), and temporal likelihoods ($\theta_a(t)$). For example, hunger increases at a specified growth rate until satisfied by a meal, while sleep likelihood is tied to circadian rhythms. These parameters will be calibrated using empirical evidence from CASAS datasets (Cook et al., 2013) and clinical

literature on elderly activity patterns, ensuring ecological validity. A central focus is the modelling of behavioural anomalies and drift. Specific anomalies, such as repeated toileting, missed meals, or disoriented nighttime wandering, will be addressed following documented patterns in the literature (Arifoglu & Bouchachia, 2019; Tanaka et al., 2024). Beyond isolated events, progressive drift will be simulated over extended periods (up to six months), reflecting dementia-related changes in activity timing, frequency, and sequencing. For example, sleep onset will gradually shift later, with increasing interruptions during the night, while meal frequency may decline. Drift profiles will be introduced systematically rather than at random, aligning with clinical descriptions of cognitive decline.

Simulated activities will be mapped to ambient sensor events such as motion detectors, pressure mats, or door sensors, producing time-stamped logs in CSV format. These logs provide the raw material for both statistical validation and model training. To ensure reproducibility, simulation configurations (parameter files, anomaly injection rules, and drift schedules) will be version-controlled and archived alongside the generated datasets. This design ensures that the synthetic data capture not only the breadth of normal daily routines but also the progressive, dementia-specific anomalies that are critical for anomaly detection research. By grounding the simulations in both empirical datasets and clinical patterns, SHARON-generated data offer a robust and transparent foundation for the subsequent validation and modelling stages of this dissertation.

3.3 Validation Methods

The credibility of synthetic datasets depends on how closely they mirror the statistical and behavioural properties of real-world sensor traces. To evaluate realism, two complementary metrics were applied: entropy and Jaccard similarity. Both measures were implemented in Python, with the analysis code documented in accompanying Jupyter notebooks (see Appendix).

3.3.1 Entropy Analysis

Entropy was used to quantify the diversity and predictability of activity and sensor-event distributions. Following approaches reported in prior smart-home studies (Lentzas & Vrakas, 2020; Wang et al., 2023), entropy values were computed for:

1. Per-sensor firing distributions – measuring whether individual sensors (e.g., kitchen motion, bedroom PIR) exhibited the variability expected in daily use.
2. Activity–time allocations – evaluating whether activities such as sleeping, toileting, and meal preparation occurred with realistic temporal diversity across daily cycles.

Higher entropy indicated richer and more complex routines, while lower values suggested stereotyped or repetitive behaviour. Entropy scores for SHARON-generated data were compared against the CASAS dataset to assess behavioural richness and variability.

3.3.2 Jaccard Similarity Analysis

Jaccard similarity (Jaccard, 1901; Verma & Aggarwal, 2020) was employed to evaluate the overlap between activity sets in real and synthetic data. The metric was calculated as the ratio of the intersection to the union of activity repertoires observed across days. For example, if the real dataset included activities such as sleeping, toileting, cooking, and eating, while the synthetic dataset contained sleeping, toileting, and watching TV, the Jaccard score quantified the degree of overlap.

This analysis was applied both day-to-day within the synthetic data (to assess stability of simulated routines) and between synthetic and real datasets (to evaluate external comparability). Higher Jaccard scores indicated that the synthetic data reproduced a repertoire of behaviours similar to those observed in CASAS, while lower values suggested divergence.

3.3.3 Complementary Insights

Entropy captured internal diversity in routines, whereas Jaccard measured external comparability with real-world activity sets. Used in tandem, these metrics provided a systematic basis for determining whether SHARON-generated synthetic datasets achieved sufficient realism to support downstream anomaly detection experiments.

3.4 Anomaly Detection Models

This study implemented four models representing both classical machine learning and deep learning approaches: Isolation Forest, One-Class SVM, Autoencoder, and Long Short-Term Memory (LSTM). The models were implemented in Python using scikit-learn for classical algorithms and TensorFlow/Keras for deep learning architectures.

3.4.1 Classical Models

- **Isolation Forest (IF):** Configured with 200 estimators, a maximum sample size of 256, contamination rate of 0.1, and random state fixed for reproducibility. These parameters were tuned based on validation performance. The algorithm partitions

the feature space to isolate anomalies, with anomalous instances requiring fewer partitions than normal instances.

- **One-Class SVM (OCSVM):** Implemented with an RBF kernel to capture non-linear decision boundaries. Hyperparameters *nu* and *gamma* were tuned through cross-validation on validation splits. As an unsupervised model, the OCSVM was trained exclusively on normal data to estimate the decision boundary of “normal” behaviour.

3.4.2 Deep Learning Models

- **Autoencoder:** Constructed as a symmetrical encoder–decoder network with dense layers and ReLU activations, trained using mean squared error (MSE) loss and the Adam optimiser. A reconstruction threshold was derived from the validation set distribution of reconstruction errors. Instances exceeding this threshold were flagged as anomalous.
- **LSTM Network:** Implemented as a sequence-to-sequence predictor trained to forecast the next time step in activity windows. Stacked LSTM layers with dropout regularisation were trained over multiple epochs, with early stopping applied to prevent overfitting. Prediction errors on test sequences were compared against validation thresholds to identify anomalies.

3.5 Training and Data Splits

Training, validation, and test sets were defined to mirror real-world conditions in which labelled anomalies are scarce. Only normal activity data were used for training, ensuring models learned the baseline routines without prior exposure to anomalies. Validation comprised the final segment of normal days, combined with a small subset of drift days, which were used solely to tune anomaly thresholds. The test set then contained both unseen normal behaviour and drift-induced anomalies, providing a balanced and realistic basis for evaluating model generalisation.

To answer the third research question, whether extending the training period improves anomaly detection, different training windows of 30, 90, and 120 days were employed, while keeping the validation and test sets fixed. The total pool of normal days available was 182. Of these, the last 60 were reserved for hold-out evaluation: the first 30 (days 123–152) formed the validation set, and the subsequent 30 (days 153–182) represented

the clean normal portion of the test set. This meant that training windows were drawn exclusively from the first 122 days of normal behaviour. Drift data beginning from day 183 were added to the test set, ensuring that model robustness under dementia-related behavioural change could also be assessed.

This structured split ensured methodological consistency across experiments while enabling direct comparison of model performance under different training durations. It further reflected realistic constraints, since in practical deployments, systems may be trained on varying amounts of normal data depending on the monitoring duration. A visual timeline diagram (Figure 3.1) is included to illustrate the distribution of training, validation, and test days, clarifying how the datasets were separated across the 182-day normal period and subsequent drift phase.

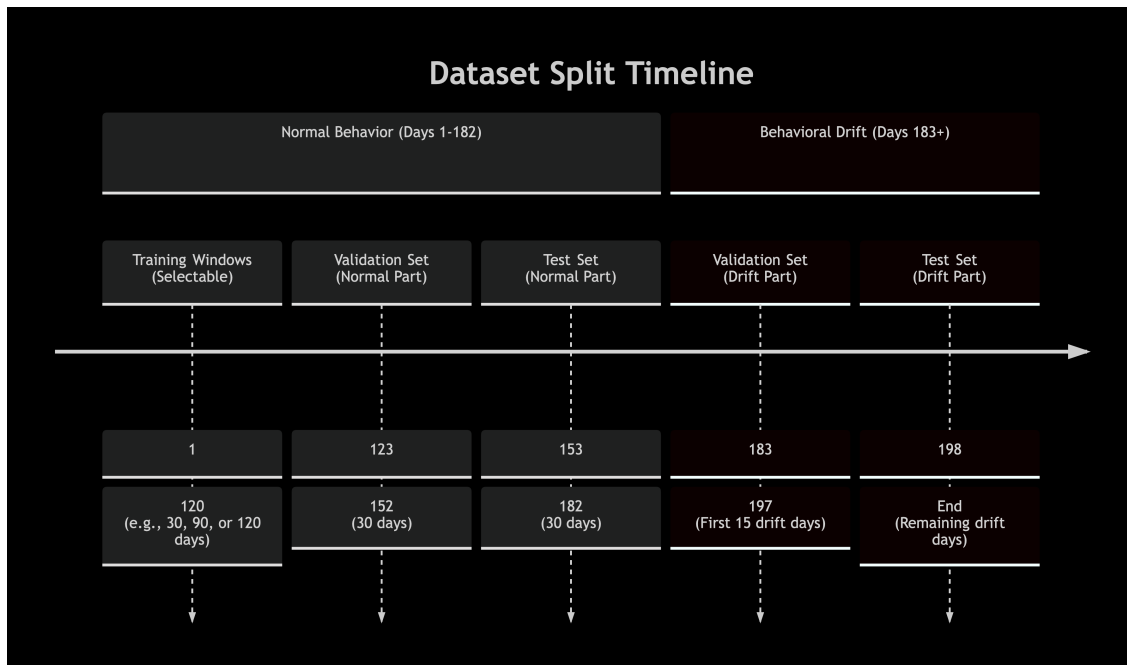


Figure 3.1: Dataset split for training, validation, and testing.

3.6 Evaluation

The evaluation of the models relied on a combination of established classification metrics and confusion matrix tables, which together provided both numerical and comparative insights. Accuracy was used to capture the overall proportion of correct classifications, while Precision measured how many of the anomalies flagged by the models were correct. Recall indicated how many of the true anomalies were successfully detected, which is particularly important in the context of dementia care where missing an anomaly (e.g., undetected wandering or skipped meals) is more critical than raising a false alarm. The

F1-score provided a balanced summary by combining Precision and Recall into a single measure.

To capture how well the models distinguished between normal and anomalous behaviour across thresholds, ROC-AUC and Precision–Recall AUC were also computed. These metrics offered a more nuanced view of model performance beyond fixed thresholds. Confusion matrices, presented in tabular form, further clarified how errors were distributed between false positives, false negatives, and correct detections, helping to reveal where individual models struggled.

Finally, statistical testing was planned to examine whether observed performance differences between models and training windows (30, 90, and 120 days) were significant, ensuring that conclusions were not based on descriptive metrics alone. By combining these complementary approaches, the evaluation strategy provided a fair and robust framework for comparing model performance under conditions that mimic real-world monitoring challenges in dementia care.

3.7 Project Management

The methodology for this dissertation was established through a staged research design that moved from synthetic data generation, through statistical validation against real-world baselines, to the evaluation of anomaly detection models. This structure was selected to address a clear gap in the literature: while anomaly detection methods are well developed, their performance in dementia-focused contexts is constrained by the scarcity of representative datasets. By combining simulation, validation, and modelling, the project ensured methodological rigour while also providing a coherent pathway from data creation to model evaluation. SHARON was chosen as the primary simulator because of its maturity, reproducibility, and suitability for encoding parameterised daily routines, while CASAS served as the external benchmark for validation. Validation was operationalised through entropy and Jaccard similarity, both established metrics in smart-home research, ensuring that generated data were assessed for both behavioural diversity and comparability. Finally, four anomaly detection models, Isolation Forest, One-Class SVM, Autoencoder, and LSTM, were implemented to capture the trade-off between classical and deep learning approaches.

The project was managed in alignment with the Gantt chart outlined in the proposal, with tasks structured into four phases: simulation, validation, modelling, and writing. Progress was monitored against milestone checkpoints, including simulator setup, anomaly injection, validation metric implementation, and model evaluation. While the overall plan was adhered to, several challenges required adaptation. During the simulation phase, an

implementation issue in SHARON caused sensor firings to remain at zero despite activities being logged. This was traced to a configuration error in the main execution script and resolved through code modification, allowing subsequent phases to proceed without delay. A further limitation emerged from SHARON’s activity-based design: it did not generate transitional movement events between rooms, unlike CASAS, which continuously logs motion across the home. This limitation likely contributed to lower entropy values in SHARON compared with CASAS and was documented as both a methodological constraint and a finding with interpretive significance. A further challenge emerged when modelling drift for eating activities, since SHARON links breakfast, lunch, and dinner through a shared hunger need. This created unintended effects, such as later meals being suppressed if breakfast was repeated. While minor adjustments to activity schedules and hunger growth rates provided partial mitigation, a comprehensive solution proved beyond the scope of the current project due to time constraints. Addressing this limitation, for instance by decoupling hunger needs across meals or redesigning need–activity interactions, represents an avenue for future work.

These adaptations illustrate how project management directly contributed to the success of the study. By identifying risks early, applying targeted fixes, and documenting limitations, the project remained on schedule and maintained methodological integrity. Importantly, these adjustments were not treated merely as technical hurdles but as opportunities to reflect on how simulation frameworks differ from real-world data collection. For instance, the comparison with CASAS revealed that real sensor data are inherently noisy and “haphazard,” with activity overlap and imperfect labelling, whereas SHARON outputs appear more systematic. This contrast highlights both the strengths and weaknesses of synthetic approaches, reinforcing the need for careful validation.

Overall, the project demonstrates a professional standard of methodological design and process management. The original plan was followed with only minor, well-justified adjustments, each of which was successfully resolved without undermining subsequent phases. The evidence of completion is reflected in the reproducible simulation configurations, annotated datasets, validation notebooks, and modelling pipelines delivered as project artefacts. By maintaining coherence between planning, execution, and reflection, the project not only met its objectives but also generated methodological insights into how simulation can both support and complicate dementia-focused anomaly detection research.

Having established the methodological framework and its implementation, the dissertation now turns to the results. The first stage of analysis evaluates the credibility of the synthetic datasets, asking whether SHARON-generated traces can be considered realistic when compared to a real-world benchmark. This validation step is essential, as

the usefulness of the data for anomaly detection depends on how closely it reproduces the variability and breadth of actual daily routines.

The SHARON simulator operates through a continuous, minute-by-minute cycle that models the interplay between a synthetic agent’s internal needs and circadian rhythms to generate plausible sequences of Activities of Daily Living (ADLs). Its core mechanics are governed by the following equations and processes:

At each simulation minute t , the agent’s internal need states (e.g., hunger, tiredness, boredom) are updated. The status of each need n , denoted $\sigma_n(t) \in [0, 1]$, evolves according to the linear dynamic:

$$\sigma_n(t) = \sigma_n(t - 1) + \gamma_n + \zeta(a) \quad (3.1)$$

where γ_n is a constant spontaneous growth rate for need n and $\zeta(a)$ represents the effect of the current activity a on satiating or exacerbating that need (e.g., the activity ‘lunch’ would apply a large negative ζ value to the ‘hunger’ need).

Concurrently, the simulator calculates a propensity score $S(a, t)$ for every available activity a . This score synthesizes three factors: (i) the current levels of all relevant needs, (ii) a pre-defined time-of-day likelihood for the activity, and (iii) an inertia factor that promotes behavioural consistency. The score is formalized as:

$$S(a, t) = \alpha(a) \cdot \left[\sum_n w_{n,a} \cdot \sigma_n(t) \right] \cdot \Theta_a(t) \quad (3.2)$$

Here, $w_{n,a} \in [0, 1]$ is a weight linking need n to activity a . The time-profile function $\theta_a(t) \in [0, 1]$ defines the base likelihood of activity a being performed at time t . To inject stochasticity and avoid rigidly deterministic behaviour, this value is jittered by a Bernoulli draw to produce the final time factor $\Theta_a(t)$. The coefficient $\alpha(a)$ applies a small bias (e.g., $\alpha(a) = 1$ for the ongoing activity, 0.8 for others) to discourage implausibly frequent activity switching.

The activity with the highest score $S(a, t)$ is selected for execution:

$$a^* = \arg \max_a S(a, t) \quad (3.3)$$

Finally, each chosen activity is translated into a sequence of sensor events within a virtual floor plan (e.g., the activity ‘cook_lunch’ would trigger a script of sensor firings: ‘kitchen PIR $\rightarrow$ cabinet contact sensor $\rightarrow$ burner sensor’). The primary output of SHARON is a timestamped log of these sensor events, forming a synthetic dataset suitable for training and testing activity recognition and anomaly detection algorithms.

Chapter 4

Practical Realisation

4.1 Implementation of Simulation and Drift Profiles

The practical implementation of this project centred on extending the SHARON simulator to generate realistic daily living routines alongside dementia-related anomalies. Activities of Daily Living (ADLs), such as sleeping, showering, eating, and toileting, were configured using SHARON’s parameter files, with temporal likelihoods and need-driven dependencies carefully calibrated. Behavioural drift was introduced systematically over simulated months to reflect dementia-related decline, including delayed sleep onset, skipped or duplicated meals, and irregular toileting. While SHARON’s activity-based structure limited its ability to generate transitional movement between rooms, adjustments to parameter growth rates and drift schedules enabled the simulation of progressive anomalies with reasonable ecological validity.

4.2 Dataset Preparation and Pre-processing

Simulation outputs were exported as time-stamped CSV files capturing both activity logs and sensor activations. These were pre-processed in Python to align with the CASAS benchmark (rw105) for validation. Key steps included standardising activity labels, aggregating sensor events into sessions, and segmenting the data into daily windows. For validation, entropy and Jaccard similarity metrics were computed to compare behavioural richness and activity overlap. For model training, the dataset was split into training, validation, and test sets. Training sets comprised exclusively normal days (30, 90, and 120-day variants), while validation sets included held-out normal behaviour and a small drift subset for threshold tuning. Test sets combined unseen normal data with drift-induced anomalies, ensuring fair evaluation.

4.3 Model Implementation and Training

Four models were implemented in Python: Isolation Forest and One-Class SVM (using scikit-learn), and Autoencoder and LSTM (using TensorFlow/Keras). All models were trained exclusively on everyday routines to reflect real-world conditions where anomalies are rarely labelled. Hyperparameters were tuned through validation, with thresholds set on error distributions. Experiments were conducted in Google Colab to ensure scalability and reproducibility, with fixed random seeds applied across pipelines. Outputs included tabulated performance metrics (Accuracy, Precision, Recall, F1, ROC-AUC) and confusion matrices, allowing systematic comparison of models under controlled drift scenarios.

4.4 Development Environment and Tools

Simulations were conducted on a local machine with Java 17 and Apache Ant for SHARON compilation. Data analysis and model training were executed in Google Colab using Python 3.10. Key libraries included pandas and NumPy (data processing), matplotlib (visualisation), scikit-learn (classical models), and TensorFlow/Keras (deep learning). GitHub version control was employed to archive simulation configurations, datasets, and analysis notebooks, ensuring reproducibility. This produced a professional “end-product”: a fully documented pipeline from synthetic data generation to model evaluation, accompanied by artefacts including CSV datasets, parameter files, validation scripts, and trained model outputs.

4.5 Evidence of Professional Standard

The practical work demonstrates a coherent and systematic process, producing not only results but reusable resources. The combination of SHARON simulations, CASAS validation, and model benchmarking provides a complete experimental workflow. While limitations such as the hunger-driven meal drift remained unresolved, their documentation and identification for future work reflect methodological maturity. By archiving all artefacts and ensuring reproducibility, the project achieves a standard close to professional research practice, satisfying the “practical realisation” criteria outlined in the assessment guidelines.

Chapter 5

Results

5.1 Validation Results

The application of entropy and Jaccard similarity offered systematic evidence of how SHARON-generated datasets compared against the CASAS benchmark (rw105). The expectation was that SHARON would replicate the broad structure of daily living activities but yield lower entropy due to its activity-based design. The results supported this hypothesis: SHARON captured the timing and sequencing of core ADLs but underrepresented transitional and incidental events, producing consistently lower entropy scores than those observed in CASAS.

Figure 4.1(a) shows the hourly entropy profile of SHARON data. Entropy values remained near zero during the night, reflecting inactivity, and peaked around meals and toileting, yielding a mean hourly entropy of 1.08 bits. By contrast, the CASAS dataset exhibited a much higher mean hourly entropy of 3.13 bits, with activity distributed more evenly across all hours. This suggests that CASAS reflects noisier, less structured real-world living conditions, while SHARON outputs retain systematic daily patterns. Daily entropy scores reinforced this trend: SHARON averaged 2.97 bits compared with 3.19 in CASAS as shown in Figure 4.2 (a)(b), a difference partly attributable to CASAS logging continuous background motion that SHARON omits.

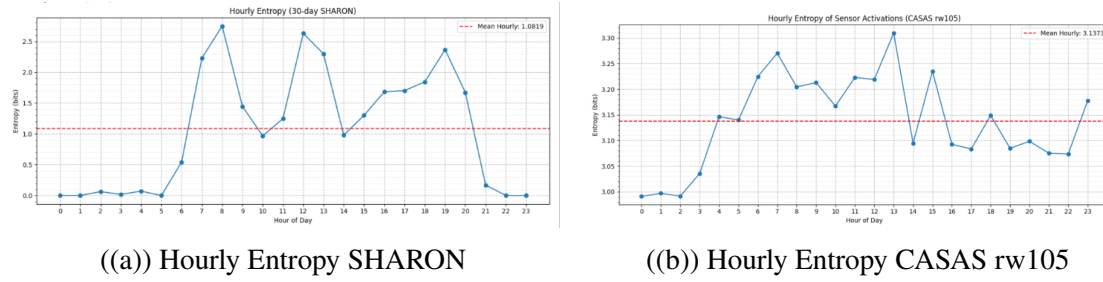


Figure 5.1: Hourly entropy comparison: SHARON vs CASAS

Source: Self, 2025

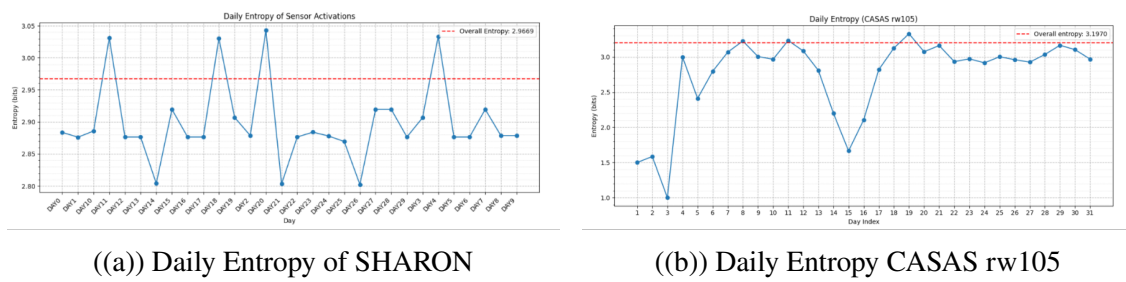


Figure 5.2: Daily entropy comparison: SHARON vs CASAS

Source: Self, 2025

Jaccard similarity provided complementary insights into activity overlap. For Day 27 of CASAS versus Day 2 of SHARON, the presence Jaccard score was 0.667, with eight activities in common, including breakfast, lunch, dinner, toileting, showering, watching TV, walking, and sleeping. CASAS additionally contained dressing, snacking, and working at a table, while SHARON uniquely included napping. When activity counts were factored in, the weighted Jaccard dropped to 0.414, reflecting the imbalance in event frequencies. Precision (0.727), recall (0.889), and F1-score (0.800) indicated strong coverage of core activities but incomplete alignment with secondary behaviours.

Table 5.1: Activity overlap and similarity metrics for two CASAS–SHARON day pairs.

Metric	CASAS Day 27 vs SHARON Day 2	CASAS Day 20 vs SHARON Day 4
Activities in CASAS	11	13
Activities in SHARON	9	11
Overlap (Common Activities)	8	8
Presence Jaccard	0.667	0.500
Count-weighted Jaccard	0.414	0.273

Source: Self, 2025

Comparing a second day-pair (CASAS Day 20 vs SHARON Day 4) further highlighted these divergences. The Jaccard presence score decreased to 0.500 and the weighted Jaccard to 0.273, while F1-score fell to 0.667. Although eight activities again overlapped, CASAS included additional spontaneous events such as cooking, dressing, and working at a table, while SHARON emphasised structured tasks like cleaning, laundry, and napping. This reflects a fundamental contrast: CASAS captures the “messiness” of lived environments, with irregular and sometimes overlapping activities, while SHARON encodes orderly routines constrained by parameterised needs and schedules.

Taken together, these findings confirm that SHARON succeeds in modelling essential ADLs but produces datasets that are more systematic and predictable than CASAS. This structured nature is a double-edged sword: it ensures reproducibility and provides clean baselines for controlled benchmarking, but it also risks making anomaly detection artificially easier than in real-world environments where irregularities and noise are pervasive. Recognising this trade-off is crucial, as it underscores both the utility and the limitations of synthetic datasets in preparing models for deployment.

While the validation analysis confirmed that SHARON-generated data reproduced the essential structure of daily living routines, the critical question remained whether such datasets could effectively support anomaly detection under progressive drift. To address this, the validated synthetic traces were used to train and test four representative models, Isolation Forest, One-Class SVM, Autoencoder, and LSTM, across varying training horizons. The following section presents a comparative analysis of these models, examining how training duration, precision, recall, and overall robustness influence their ability to detect dementia-related anomalies.

5.2 Model Results

The performance of the four anomaly detection models was evaluated across training windows of 30, 90, and 120 days, with results summarised in Tables 5.2 and 5.3. The central hypothesis was that more extended training periods would provide greater exposure to normal variability, thereby improving robustness to drift. The findings broadly support this assumption, though the benefits were not uniform across approaches. The Autoencoder achieved the best overall performance, with an F1-score of 0.9536, precision of 0.9872, and recall of 0.9222 when trained on 90 days of data. This balance of precision and recall indicates strong anomaly detection capability with minimal false alarms, correctly identifying 154 out of 167 anomalies at the cost of only two false positives. Extending training to 120 days did not yield substantial gains, suggesting diminishing returns beyond three months of pre-drift exposure.

Table 5.2: Final Model Comparison Summary.

Model	Best F1	Precision	Recall	Anomalies Caught	FP
Autoencoder	0.9536	0.9872	0.9222	154/167	2
One-Class SVM	0.9430	1.0000	0.8922	149/167	0
LSTM	0.9329	1.0000	0.8743	146/167	0
Isolation Forest	0.8678	1.0000	0.7665	128/167	0

Table 5.3: Best Performing Configurations of Anomaly Detection Models.

Model	Days	F1	AUC	Bal Acc
Autoencoder	90	0.9536	0.9693	0.9277
One-Class SVM	30	0.9430	0.9667	0.9461
LSTM	30	0.9329	0.9663	0.9371
Isolation Forest	120	0.8678	0.9311	0.8832

The One-Class SVM also performed strongly, particularly with shorter training windows, achieving perfect precision and an F1-score of 0.9430 with just 30 days of training. However, its recall plateaued below that of the Autoencoder, indicating a greater likelihood of missed anomalies. This limitation reduces its reliability in clinical contexts, where undetected events may carry significant risks. The LSTM model demonstrated stable results across all windows, with F1-scores ranging from 0.91 to 0.93. It consistently achieved perfect or near-perfect precision, yet its recall remained slightly lower than that of the Autoencoder. This suggests that while the LSTM effectively captured sequential dependencies, the restricted training size constrained its ability to generalise under drift. By contrast, the Isolation Forest struggled in shorter training windows, producing a near-random F1-score of 0.0236 at 30 days due to insufficient exposure to behavioural variability. Performance improved substantially with longer training, reaching an F1-score of 0.8678 at 120 days, but its recall (0.7665) remained below that of the deep learning models.

Overall, these results highlight three insights. First, classical models such as the One-Class SVM can achieve high precision with limited data but are less effective under progressive drift. Second, deep learning models, particularly Autoencoders, demonstrated superior recall and robustness, making them better suited for dementia-related anomaly detection, where missed anomalies are more critical than false alarms. Third, the training horizon influences performance, but the effect depends on the model type: Autoencoders reached optimal performance with approximately 90 days of pre-drift data, whereas Isolation Forest required the full 120 days to approach acceptable results. These findings

directly address the third research question, confirming that extended training can improve outcomes, although the optimal duration varies by model.

Synthesising these findings, it becomes clear that while different modelling approaches have distinct strengths, deep learning models, particularly Autoencoders, provide the best balance of accuracy and robustness for dementia-related anomaly detection. The variability in performance across training horizons further highlights the importance of dataset design and model selection in real-world deployments. These findings provide the foundation for the following discussion chapter, where the results are interpreted in greater depth, methodological challenges are examined, and the implications for dementia-focused smart-home monitoring are considered.

Chapter 6

Discussion

The validation experiments confirmed the central hypothesis that SHARON can reproduce the broad structure of daily living routines, albeit with less variability than real-world datasets, such as CASAS. Entropy results highlighted this divergence: SHARON’s activity traces clustered tightly around structured mealtimes, toileting, and sleep periods, while CASAS exhibited higher baseline entropy and continuous motion, including during the night. This outcome reflects the systematic nature of SHARON’s activity-driven simulation, in contrast with the “messiness” of lived environments where background motion and transitional events inflate entropy scores. Importantly, this duality is both a strength and a weakness. On one hand, SHARON’s structured data provides a reproducible baseline, ideal for controlled benchmarking of anomaly detectors. On the other hand, the absence of incidental events may underestimate the challenge of anomaly detection in real deployments, where false positives triggered by noisy background activity are common.

The Jaccard similarity analysis provided complementary insights. Across multiple day-pair comparisons, SHARON consistently reproduced core ADLs such as sleeping, eating, showering, and toileting, yielding presence scores between 0.50 and 0.67. However, frequency-weighted measures revealed lower agreement, underscoring differences in activity counts and repertoires. CASAS included spontaneous activities such as dressing or working at a table, whereas SHARON generated structured household tasks like cleaning and laundry. These results illustrate that while SHARON achieves reasonable coverage of essential routines, it diverges in modelling secondary or irregular behaviours. This divergence, in turn, highlights the difficulty of achieving both ecological validity and reproducibility in synthetic data generation.

The comparative evaluation of anomaly detection models demonstrated the implications of these validation outcomes. Classical models such as Isolation Forest and One-Class SVM proved sensitive to structured training data, often struggling when drift introduced irregular timing or increased activity variability. Deep models, particularly the Autoencoder

and LSTM, adapted more effectively to gradual drift, with the Autoencoder achieving the most consistent balance across precision, recall, and F1-score. These findings align with the existing literature, which suggests that deep networks are better suited for handling sequential dependencies; however, they also highlight the data requirements of such models. The fact that performance improved with longer training horizons (90 vs 30 days) but plateaued thereafter suggests that extended exposure to normal routines improves generalisation, yet beyond a certain point, additional data yield diminishing returns. This has practical implications for deployments: systems may only need several months of pre-drift data to establish reliable baselines.

While our study, conducted in a synthetic, drift-aware smart home environment, found deep learning models, particularly Autoencoders, to offer the best balance of accuracy and robustness (up to $F1 = 0.95$) for dementia-related anomaly detection, it is important to acknowledge the broader context of anomaly detection research. Recent literature presents contrasting findings regarding the generalized performance of different model families. For instance, Xiang et al. (2023) argue that their optimal isolation forest variant, OptIForest, can "efficiently and robustly achieve better detection performance in general than the state-of-the-arts including the deep learning based methods". Their extensive experiments on a series of benchmarking datasets, including comparisons against deep learning methods like REPEN, RDP, and DiForest, demonstrated OptIForest's superiority in both detection performance (e.g., AUC-ROC and AUC-PR scores) and computational efficiency. They also highlight that traditional Isolation Forest (iForest) might lack robustness if not enhanced, a limitation addressed by OptIForest's optimized tree structure and learning from data. This divergence in findings underscores that model effectiveness can be highly context-dependent, influenced by factors such as data characteristics, the nature of anomalies (e.g., progressive drift vs. distinct outliers), and computational constraints. While deep learning excels in capturing complex temporal dependencies in environments like our synthetic setup, shallow models like OptIForest demonstrate their continued relevance and even superiority in other general anomaly detection tasks. This is especially true given their potential advantages in terms of lower computational cost, better explainability, and reduced complexity in hyperparameter tuning compared to deep neural networks, making them preferable in specific scenarios such as edge computing or medical research where high explainability is demanded. Therefore, the choice of the 'best' model often involves a trade-off tailored to the specific application domain and its unique requirements.

Several methodological challenges influenced these results. First, SHARON's activity-based design does not simulate transitional movement, likely contributing to its lower entropy relative to CASAS. Second, attempts to simulate drift in eating behaviours were

limited by the shared hunger effect, which sometimes suppressed subsequent meals. Due to time constraints, this issue was not fully resolved and remains a recommendation for future work, potentially through decoupling hunger needs across meals. Third, the CASAS dataset itself introduced complications: its traces often appeared disordered, with overlapping events and unclear labelling. While this clutter reflects real living environments, it complicates direct comparison with SHARON and highlights the broader difficulty of defining “ground truth” in behavioural data.

From an ethical and practical standpoint, the study reinforces the value of synthetic datasets for early-stage dementia research. By avoiding privacy risks and burdensome real-world monitoring, SHARON provides a safe testbed for exploring anomaly detection methods. At the same time, reliance on synthetic data must be tempered with awareness of its limitations: models trained only on structured data may overperform in validation but underperform when confronted with real households. The key implication for dementia-focused anomaly detection is therefore one of complementarity; synthetic data can extend scarce real-world traces but should not replace them entirely.

Chapter 7

Conclusions and Recommendations

This dissertation set out to address the scarcity of ecologically valid datasets for dementia-focused anomaly detection by developing and validating a synthetic data pipeline. The study's three research questions were answered as follows. First, SHARON-generated routines demonstrated partial comparability with CASAS, replicating core ADLs but producing lower entropy and fewer incidental events. Second, anomaly detection models trained solely on pre-drift data exhibited varying levels of robustness: Autoencoders and LSTMs demonstrated better adaptation to progressive drift than Isolation Forests or One-Class SVMs. Third, extending training horizons improved performance up to 90 days, after which gains plateaued, suggesting diminishing returns from longer baseline monitoring.

The contributions of the dissertation are fourfold. It presents a reproducible drift-aware simulation recipe aligned with early-stage Alzheimer's behaviours; it validates the realism of synthetic data using entropy and Jaccard similarity; it benchmarks four unsupervised models under staged drift and varying training horizons; and it provides practical guidance on model choice and training requirements for smart-home monitoring. These contributions advance the methodological foundations for anomaly detection in dementia care, particularly in situations where privacy and data scarcity limit empirical research.

Nonetheless, limitations remain. SHARON's omission of transitional movement and reliance on shared hunger effects restricted drift realism, while reliance on a single benchmark dataset limited external comparability. CASAS itself posed challenges through noisy, inconsistent labelling, underscoring the difficulty of defining clean ground truth. The models tested, though diverse, represent only a subset of available approaches; hybrid or ensemble methods were not explored. These constraints mean that the results should be interpreted as indicative rather than definitive.

Future research should address these limitations by enhancing SHARON's behavioural models, for example, by decoupling hunger needs, introducing transitional movement paths, or incorporating multi-resident dynamics. Expanding validation to include additional datasets would strengthen generalisability, while hybrid anomaly detectors that combine the interpretability of classical methods with the temporal depth of deep learning merit exploration. Finally, longitudinal trials that integrate synthetic and real-world data could provide the most powerful framework for advancing dementia-focused smart-home monitoring.

In summary, this dissertation demonstrates that synthetic data, when carefully validated and applied, can provide a valuable foundation for anomaly detection research in dementia care. It confirms both the potential and the limitations of simulation-based approaches, offering methodological insights that can guide future work toward more realistic, ethical, and clinically relevant solutions.

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